Lecture 12 (Nernst)

Thursday, 16 February 2023 1:33 PM

$$\sum_j s_{ij} M_j = n e^-$$

Nernst Equation

$$U_i - U^\circ_{H_2/H^+} = \left(U_i^\circ - U^\circ_{H_2/H^+}\right) - \frac{RT}{F}\left(\sum_j \frac{s_{ij}}{n} \ln c_{ij}^*\right)$$

(s_{ij}: stoichiometric coefficient; n: e^- mol of product)

SHE SHE

$c_{ij}^* = c_{ij}$ for solute

$c_{ji}^* = P_{ij}$ for gas

Reversible Hydrogen Electrode

It is more convinient to reference a hydrogen electrode $\neq$ pH = 0

↳ eg: V_{cell} for H_2 fuel cell/water electrolyzer does not depend on H^+ conc. (pH)

But V_{red} & V_{ox} are both dependent on pH (i.e. the half reactions)

↳ If we ref a single half reaction to SHE, it would appear that a large potential is required @ pH > 0

Is E_{OC} @ equilibrium? i.e. V_{eq} value

To avoid confusion, we employ a RHE which is an SHE w/ a nernstian correction for pH shift

$$H_2 \rightleftharpoons 2H^+ + 2e^-$$

$n_e = 2H^+ + 2e^-$

Nernst: $V_{H_2/H^+} = \left(\cancel{V^\circ_{H_2/H^+}} - \cancel{V^\circ_{H_2/H^+}}^{\,0}\right) - \frac{RT}{2F} \ln\left(\frac{P_{H_2}}{C_{H^+}^2}\right)$; $pH = -\log_{10} C_{H^+}$

$$= \frac{RT}{F} \ln C_{H^+}$$

$$= -\frac{RT}{F} \ln\left(10^{-pH}\right)$$

$$\boxed{V_{H_2/H^+} = -0.0591\ V \text{ vs RHE}}$$